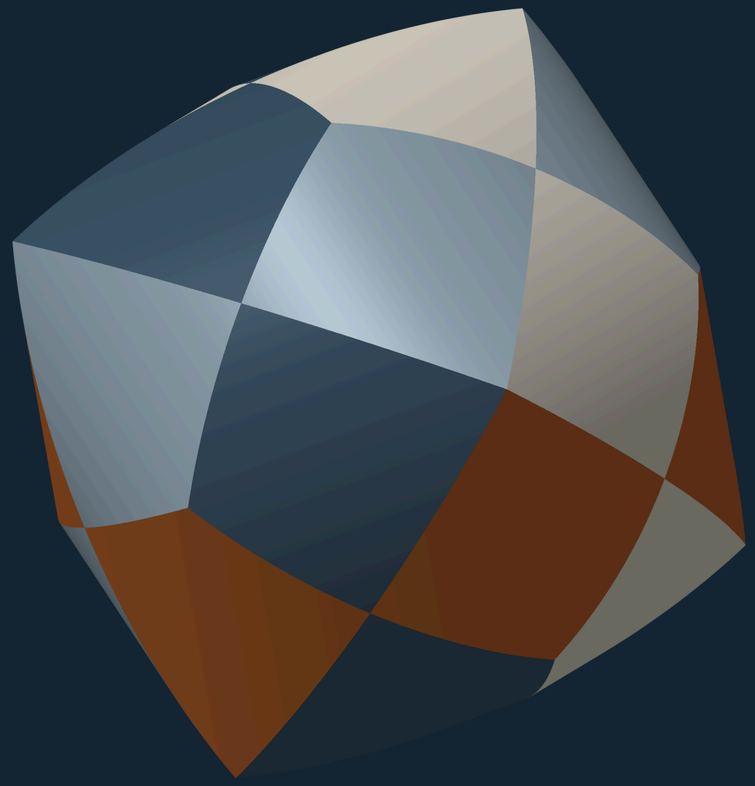
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NONLINEAR DYNAMICS LABORATORY



NCVD: an exact characteristic value decomposition

Toolbox for Python and MATLAB

github.com/nld-uri/ncvd-toolbox

Nonlinear Dynamics Laboratory · University of Rhode Island

One state record, two identities, both exact

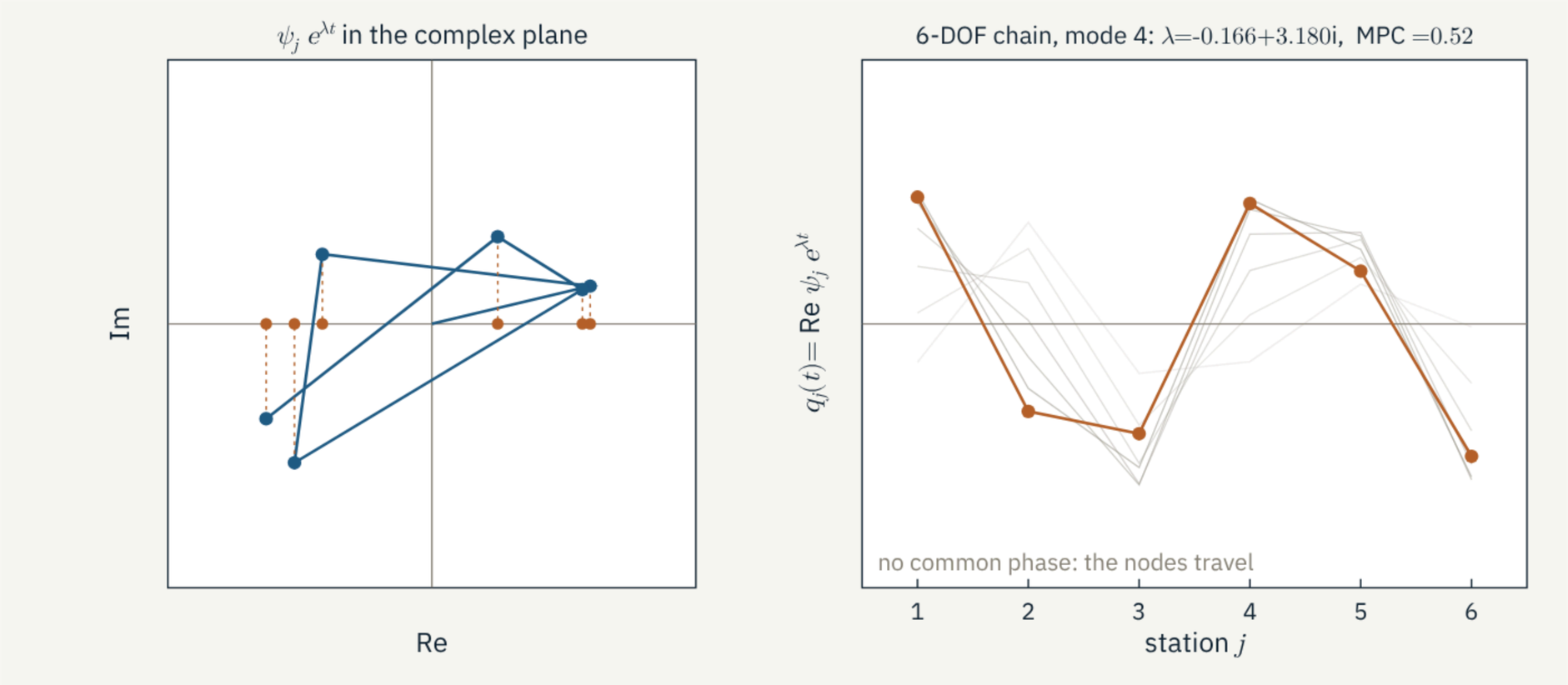
$$\mathbf{X}_1 = [\mathbf{q},\ \dot{\mathbf{q}}] = \mathbf{U}\boldsymbol{\Phi}^{\mathsf{H}}$$

$$\mathbf{X}_2 = [\dot{\mathbf{q}},\ \ddot{\mathbf{q}}] = \mathbf{U}\boldsymbol{\Lambda}\boldsymbol{\Phi}^{\mathsf{H}} + \mathbf{R}, \qquad \boldsymbol{\Lambda} = \mathrm{diag}(\lambda_i)$$

- $\mathbf{U}$: temporal basis. $\boldsymbol{\Phi}$: characteristic shapes. λ_i: characteristic values (poles).
- $\mathbf{R}$: what the diagonal form cannot represent: noise, unsupplied forcing, nonlinearity.
- Linear, noise-free record: $\mathbf{R} = \mathbf{0}$ to machine precision, and λ_i belongs to column i of $\boldsymbol{\Phi}$ by construction.

GEOMETRY

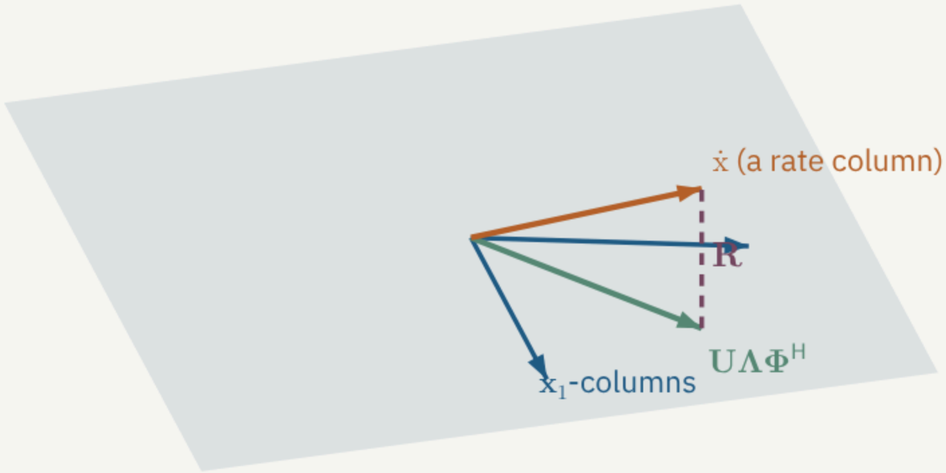
A complex mode is a rotating, shrinking polygon



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GEOMETRY

The residual is what no linear model of the record explains



$\mathbf{X}_1 = \mathbf{U}\mathbf{\Phi}^{\mathsf{H}}$ exact

$\mathbf{X}_2 = \mathbf{U}\mathbf{\Lambda}\mathbf{\Phi}^{\mathsf{H}} + \mathbf{R}$

$\mathbf{R} = (\mathbf{I} - \mathbf{X}_1\mathbf{X}_1^{\dagger})\mathbf{X}_2 \perp \operatorname{span}\mathbf{X}_1$

linear, noise-free: $\mathbf{R} = \mathbf{0}$

nonlinear window: $\|\mathbf{R}\| / \|\mathbf{X}_2\|$

is what no linear model of the window explains

01

The readout



The published GCVD reads poles from the diagonal of the coupling matrix

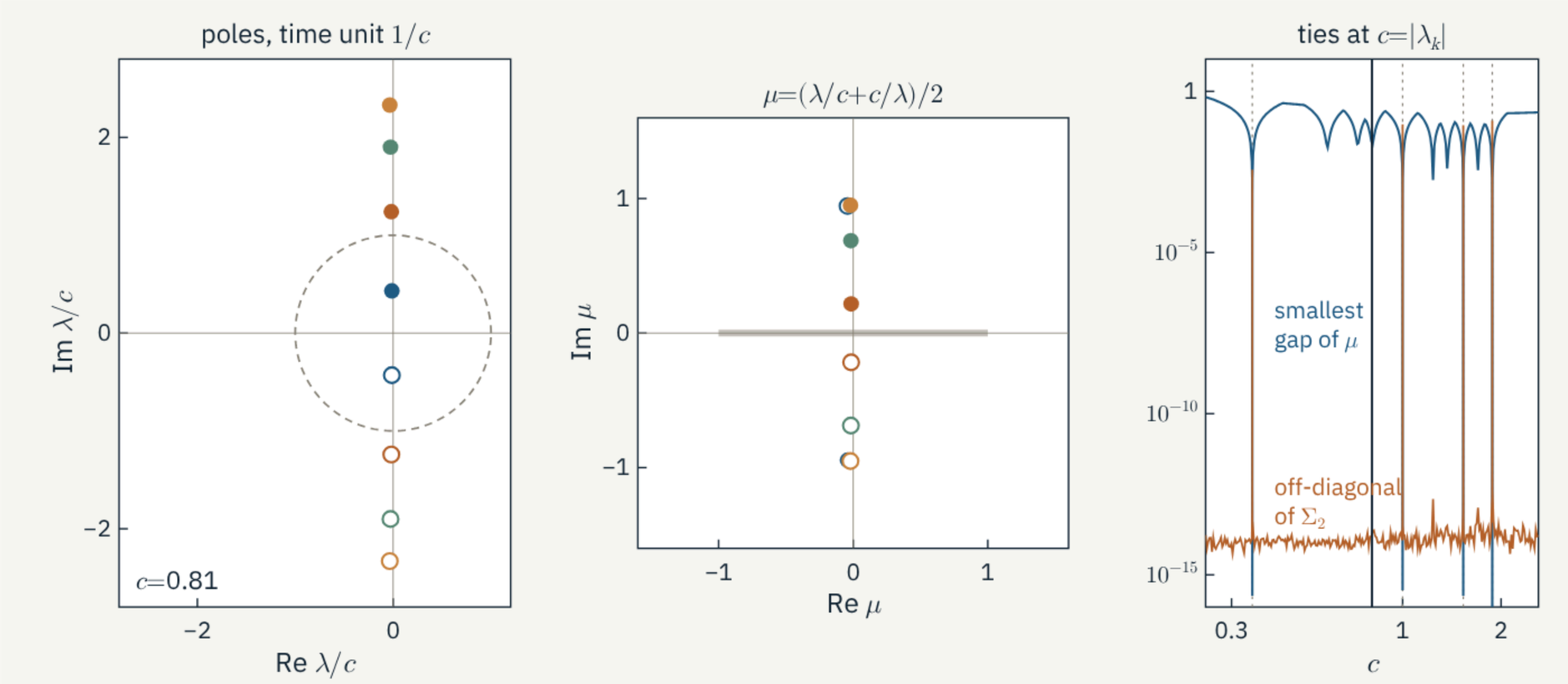
$$\mathbf{Q} = \tfrac{1}{4}\left(\mathbf{K}_{21}\mathbf{K}_{11}^{-1} + \mathbf{K}_{22}\mathbf{K}_{12}^{-1} + \mathbf{K}_{11}\mathbf{K}_{21}^{-1} + \mathbf{K}_{12}\mathbf{K}_{22}^{-1}\right), \quad \mathbf{Q}\mathbf{V} = \mathbf{V}\mathbf{M}$$

$$\boldsymbol{\Xi} = \mathbf{Y}_1\mathbf{V}^{-\mathsf{H}}, \qquad \boldsymbol{\Sigma}_2 = \mathbf{V}^{\mathsf{H}}\mathbf{B}\,\mathbf{V}^{-\mathsf{H}}, \qquad \mathbf{B} = \mathbf{Y}_1^{\dagger}\mathbf{Y}_2$$

published: $r_i = (\boldsymbol{\Sigma}_2)_{ii}$, shapes $\mathbf{V}$

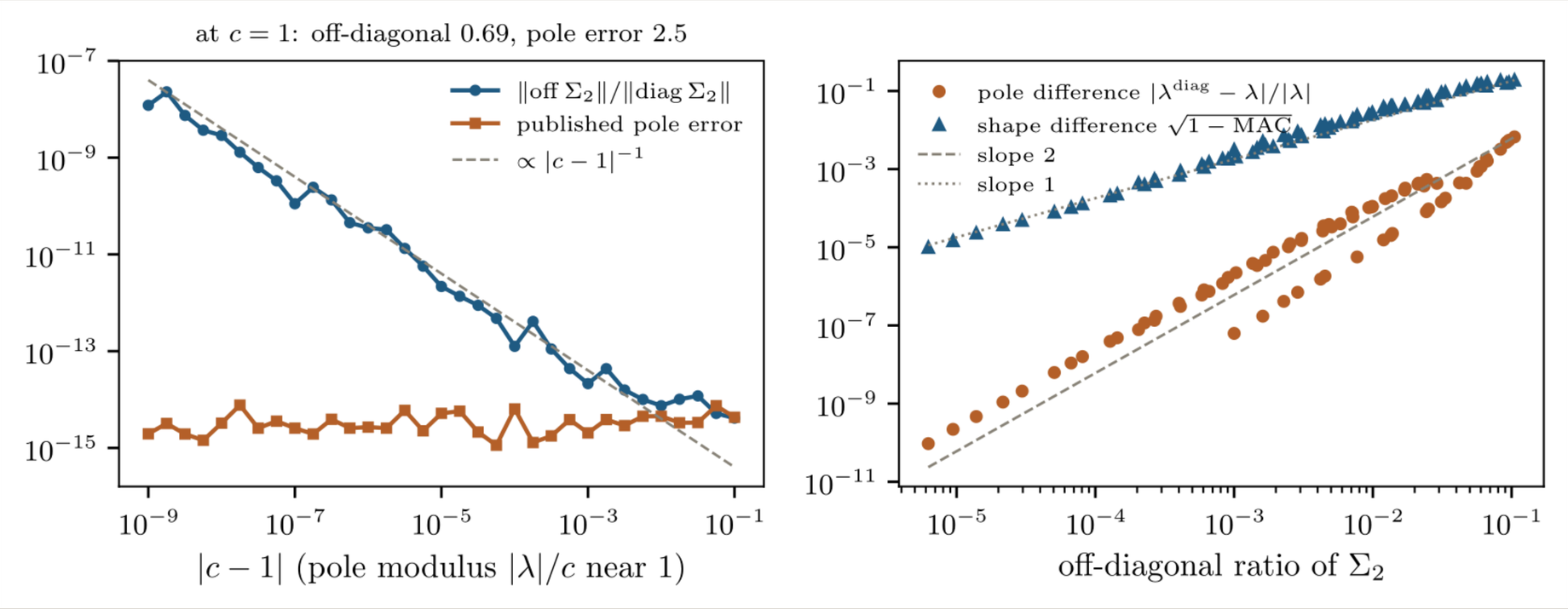
- Exact only when $\mathbf{V}$ diagonalises $\boldsymbol{\Sigma}_2$, that is, when $\mathbf{V}$ are eigenvectors of $\mathbf{B}^{\mathsf{H}}$.
- True for the CVD and “forced” quotients on any data, and for the balanced quotient on exact linear data.
- Not true for averaged ensemble quotients, nonlinear windows, or exact Joukowski ties.

The balanced quotient folds the pole plane at the unit circle



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Off the diagonal, poles err at second order and shapes at first order



Left: exact tie at $\omega = 1$. Right: windows of a Duffing decay; fitted slopes 1.74 (poles) and 1.05 (shapes).

Rotating by the eigenvectors of the coupling matrix makes both identities exact

$$\boldsymbol{\Sigma}_2 = \mathbf{T}\boldsymbol{\Lambda}\mathbf{T}^{-1}, \qquad \mathbf{U} = \boldsymbol{\Xi}\,\mathbf{T}, \qquad \boldsymbol{\Phi} = \mathbf{V}\,\mathbf{T}^{-\mathsf{H}}$$

$$\mathbf{E} = \mathbf{V}^{-\mathsf{H}}\mathbf{T}: \qquad \mathbf{B}\mathbf{E} = \mathbf{E}\boldsymbol{\Lambda}, \qquad \mathbf{U} = \mathbf{Y}_1\mathbf{E}, \qquad \boldsymbol{\Phi} = \mathbf{E}^{-\mathsf{H}}, \qquad \mathbf{R} = (\mathbf{I} - \mathbf{Y}_1\mathbf{Y}_1^{\dagger})\mathbf{Y}_2$$

- Characteristic values are the eigenvalues of the least-squares operator: the state-variable decomposition with a least-squares rate model.
- The balanced quotient fixes only the normalisation and the conditioning of the intermediate steps.
- No pole-to-shape assignment step is needed.

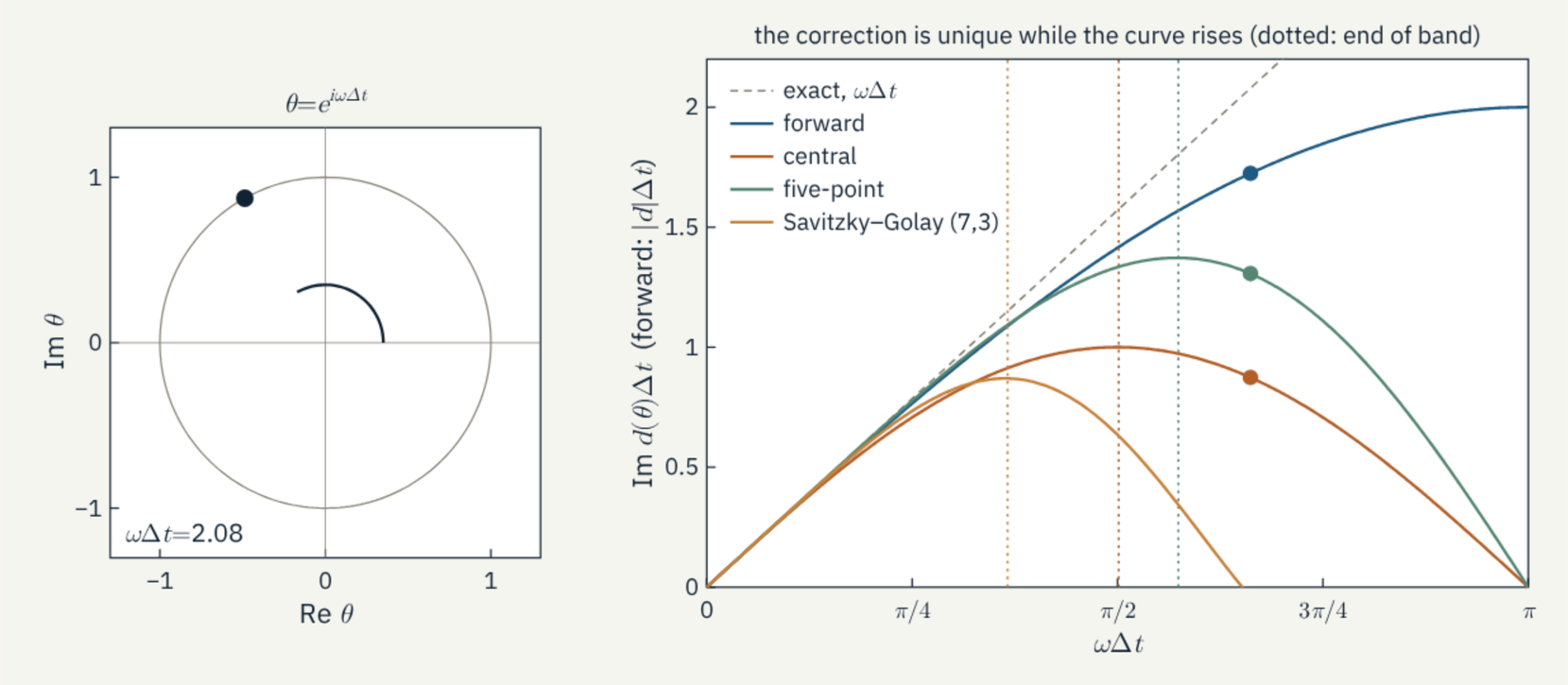
02

Practical data



PRACTICAL DATA

Any FIR derivative stencil can be inverted inside its band



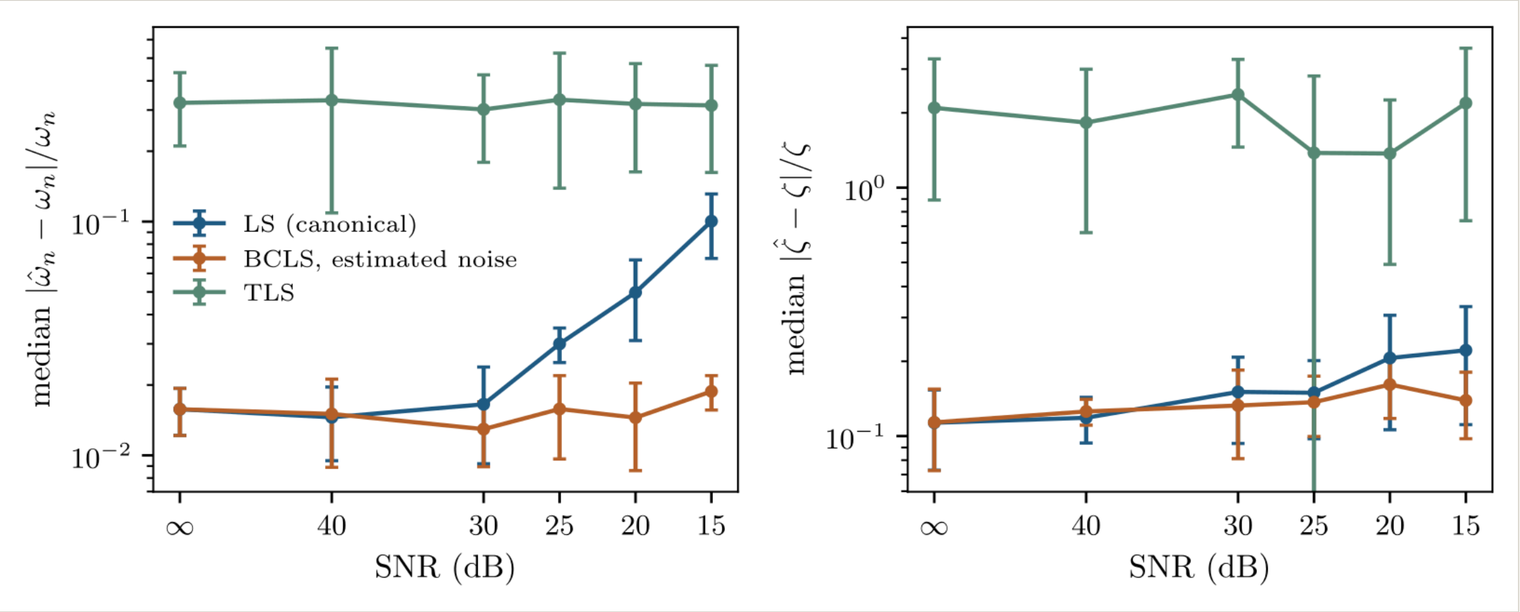
$$\tilde{\lambda} = d(e^{\lambda\Delta t})$$
$$d(\theta) = f_s \sum_j c_j \theta^j$$
$$\lambda = f_s \log\theta$$

Machine precision for forward, central, five-point and Savitzky–Golay rates.

Invertible bands: $\omega\Delta t < \pi,\ \pi/2,\ 1.80,\ 1.14.$

PRACTICAL DATA

Bias-compensated least squares removes the noise bias in frequency



$$\mathbf{B} = (\mathbf{K}_{11} - m\,\mathbf{N})^{-1}\,\mathbf{K}_{12}$$

$\mathbf{N}$: white sensor-noise variances, estimated from the flat spectrum near Nyquist.

At 15 dB: frequency error 10 % (LS) against 1.9 % (BCLS).

TLS fails under unmeasured forcing.

PRACTICAL DATA

Measured forcing and many records need no extra model

Measured forcing

- Project the input record out of both blocks.
- No mass or input matrix needed.
- Pole error at machine precision with the forcing, 0.048 rad/s without.

$$\mathbf{X}_2 = \mathbf{X}_1\mathbf{B} + \mathbf{F}\mathbf{G} + \mathbf{R}$$

Ensembles

- Stack the rows of every record and decompose once.
- Any sampling rates and lengths: the state-rate relation is continuous-time.
- 16 trials: frequency error from 1.5 % to 0.34 %.

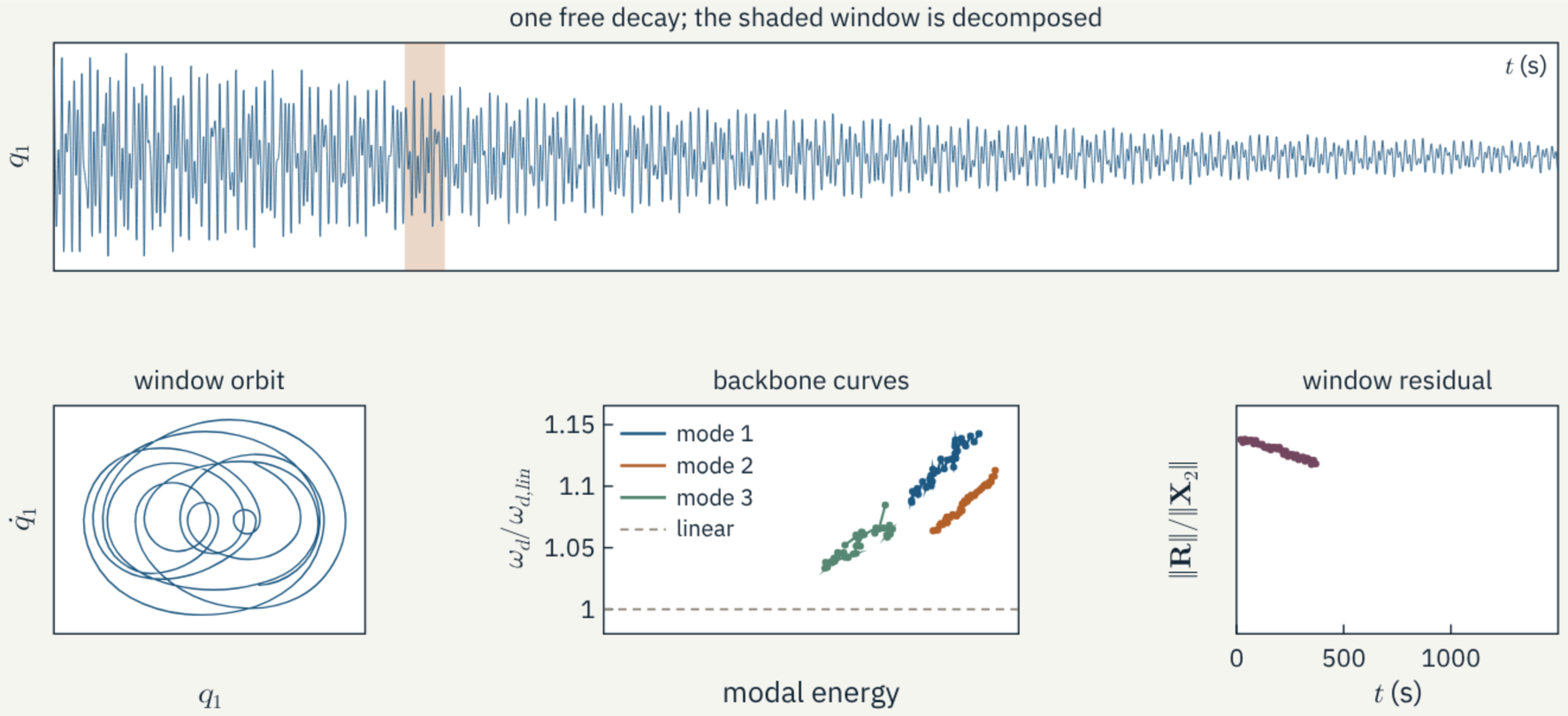
$$\begin{bmatrix} \mathbf{X}_2^{(1)} \\ \vdots \\ \mathbf{X}_2^{(N)} \end{bmatrix} \approx \begin{bmatrix} \mathbf{X}_1^{(1)} \\ \vdots \\ \mathbf{X}_1^{(N)} \end{bmatrix} \mathbf{B}$$

03

Nonlinear systems



Sliding windows trace frequency-energy dependence from one decay



Each window is an equivalent-linear fit, and it needs the right data

- Excite every resonance you track; check the singular values of each window.
- Span at least one period of the slowest mode; a dozen is comfortable.
- A weak mode can lose its pole slot to a superharmonic of a strong one.
- Validate every pole against the measured response frequency.
- Trust frequencies and shapes; single-window damping scatters.
- Shape-derived ROM quantities change at first order with the readout: do not mix readouts.

The same calls in Python and MATLAB

Python

```
import ncvd as nc
rec = nc.ncvd(d, v, a)
mp = nc.modal_parameters(rec.lam)
tr = nc.track(d, v, a, fs=fs,
length=800, step=400, M=M)
```

pytest: 28 tests

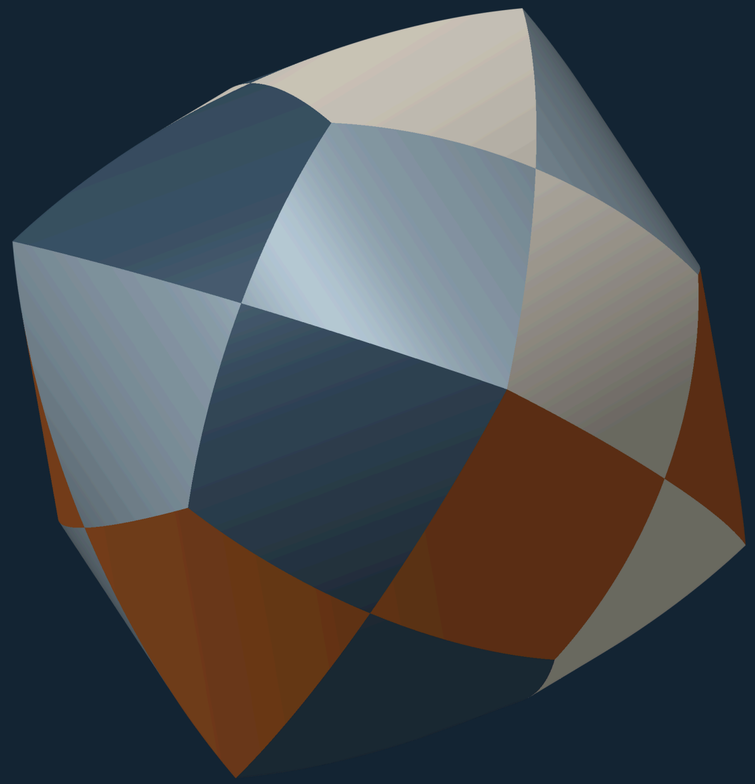
MATLAB

```
rec = ncvd.decompose(d, v, a);
mp = ncvd.modalParameters(rec.lam);
tr = ncvd.track(d, v, a, Fs=fs, ...
Length=800, Step=400, M=M);
```

runtests: 13 tests, Python cross-checks

When amplitude changes the story

In nonlinear structures, amplitude changes frequency, mode shape and stability.



github.com/nld-uri/ncvd-toolbox

web.uri.edu/nld/ncvd-toolbox

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